

IMPERIAL

Bayesian Phylolinguistic Inference under Cognate Dependence

A Study of Model Misspecification

Yong Sheng James Tay

15/06/2026

Background

Goal: Reconstruct language tree T^* from data D using Bayesian phylogenetic analysis

Language	'to laugh'		'wing'	
	Word	Class	Word	Class
Fijian	dredre	tl-A	taba-na	w-A
Marquesan	kata	tl-B	peheu	w-B
Hawaiian	'aka	tl-B	'heu	w-B
Maori	kata	tl-B	parirau	w-C
Tahitian	'ata	tl-B	pererau	w-C

Language	'to laugh'		'wing'		
	A	B	A	B	C
Fijian	1	0	1	0	0
Marquesan	0	1	0	1	0
Hawaiian	0	1	0	1	0
Maori	0	1	0	0	1
Tahitian	0	1	0	0	1

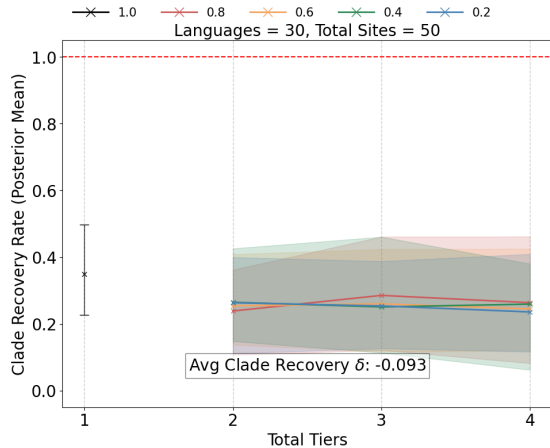
Table: Binary encoding of cognacy classes from five Oceanic languages. Adapted from Hoffman et al. (2021).

Problematic Assumption: Conditional Site Independence

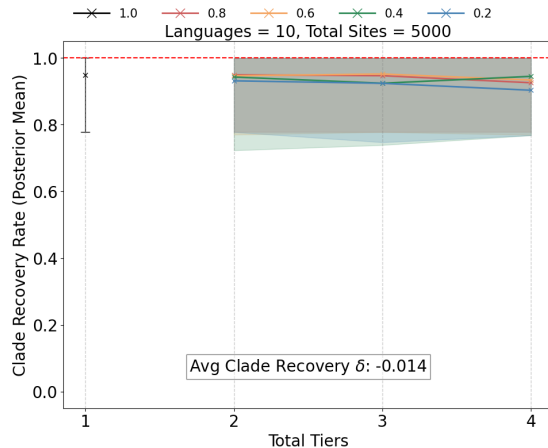
Results from Experiments

Experiment 1 (Unknown Dependency Structure)

Small Dataset



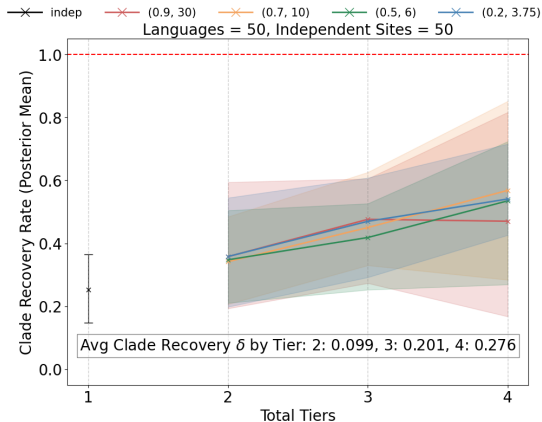
Large Dataset



Results from Experiments

Experiment 2 (Additional Dependent Sites)

Small Dataset



Large Dataset

